

PyGears, the open source chip design language based on Python released

An open source chip design language based on Python and optimised for machine learning techniques has been released by a consortium of researchers in the United States.

PyGears is a new hardware description language developed by a group at the University of California (UCLA) in the United States. It is based on the concept of reusable components and high-level Python constructs in agile chip design.

The UCLA Samueli School of Engineering claims that by employing it in the VLSI Signal Processing course, it has begun to develop a community that observes hardware through software paradigms, with the goal of expediting hardware design and verification by using more agile and composable modules.



“Chip design is getting more complex, with increasing costs and longer development cycles,” stated Dr Dejan Markovic, electrical and computer engineering professor at UCLA. “PyGears addresses these challenges by adopting a modular build of hardware out of reusable components, which are built by a community of like-minded contributors. This works very well for small and large engineering teams alike. Hardware design and verification is based on the Python environment, reducing the barriers between software and hardware domains. The vision is to enable software people to code hardware.”



malware’s creators to the United States to face charges.

Trickbot, which is based on the previous Dyre malware, is one of the world’s most hazardous botnets and is frequently used to transmit ransomware payloads. Microsoft security professionals worked with law enforcement agencies throughout the world to shut down Trickbot’s infrastructure, resulting in the arrest and extradition of some of the

Firefly introduces ValidIaC

Cloud asset management solution Firefly has announced the release of a new open source solution called ValidIaC, which enables DevOps and cloud teams to validate, secure, and map Infrastructure-as-Code (IaC) with an all-in-one browser based tool.

Inspired by Komodor’s open source project ValidKube, which combines the best open source tools to help ensure Kubernetes YAML best practices, Firefly has built a sister project for the purpose of Infrastructure-as-Code.

ValidIaC combines several open source projects and tools to allow developers to check their Terraform HCL code without changing context and from within the same workflow.



“Infrastructure-as-Code is changing the way modern organizations are managing their infrastructure. With ValidIaC, developers can move faster but still have confidence when it comes to security, compliance, cost, and reliability,” said, Ido Neeman, CEO at Firefly.

ValidIaC is an open source tool, so community contributors can add more tools or capabilities that will help them and the community. Firefly announced that it’ll soon introduce support to more Infrastructure-as-Code tools, such as Pulumi, Cloudformation, etc.

Zenoss discontinues Zenoss Community Edition

Zenoss Inc., has announced that it is sunsetting Zenoss Community Edition (previously called Zenoss Core), a free, on-prem monitoring tool the company made available for over 15 years. Since the launch of Zenoss Cloud in 2018, which saw a 202 per cent increase in annual recurring revenue over the past two years, the focus has gradually transitioned from crowdsourced development of extensions for Zenoss Community Edition to community development of cloud tools.

Zenoss Community Edition version 1.0 was released in November 2006. Since then, the product has been downloaded millions of times and became the *de facto* open source monitoring tool for companies of all sizes across all industries.

Open source trends for 2022 in Southeast Asia

Ashnik, an open source solutions provider in Southeast Asia and India, recently released its survey with participation of over 160 cross-industry organisations from the region. The survey captures the growing role of open source, as well as the key digital transformation initiatives and technology trends of 2022.